EKS Cluster Setup Guide

This document provides a comprehensive guide for setting up an Amazon EKS cluster with monitoring tools including Prometheus and Grafana. The setup includes ALB Controller, EBS CSI driver, and proper ingress configuration.

1. Create Your EKS Cluster

```
eksctl create cluster \
  --name devops-monitoring \
  --version 1.32 \
  --region us-east-1 \
  --nodegroup-name linux-nodes \
  --node-type t3.medium \
  --nodes 3 \
  --nodes-min 1 \
  --nodes-max 5 \
  --managed
```

2. Environment Setup

```
# 1. Set cluster name
export cluster_name=devops-monitoring

# 2. Get OIDC ID (optional, for advanced scripting)
oidc_id=$(aws eks describe-cluster --name $cluster_name --region us-east-1 --query "cluster.
echo "OIDC ID: $oidc_id"

# 3. Associate OIDC provider (REQUIRED for addons to work properly)
eksctl utils associate-iam-oidc-provider --cluster $cluster_name --region us-east-1 --approv
```

3. ALB Controller Setup

```
# Download IAM policy
curl -O https://raw.githubusercontent.com/kubernetes-sigs/aws-load-balancer-controller/v2.11

# Create IAM Policy
aws iam create-policy --policy-name AWSLoadBalancerControllerIAMPolicy --policy-document fil

# Create IAM Role
eksctl create iamserviceaccount \
  --cluster=devops-monitoring \
  --region=us-east-1 \
  --namespace=kube-system \
  --name=aws-load-balancer-controller \
  --role-name AmazonEKSLoadBalancerControllerRole \
  --attach-policy-arn=arn:aws:iam::264152483755:policy/AWSLoadBalancerControllerIAMPolicy \
  --approve

# Add helm repo
helm repo add eks https://aws.github.io/eks-charts

# Update the repo
helm repo update eks

# Install aws-alb-ingress-controller
helm install aws-load-balancer-controller eks/aws-load-balancer-controller \
  -n kube-system \
  --set clusterName=devops-monitoring \
  --set serviceAccount.create=false \
```

```
  --set serviceAccount.name=aws-load-balancer-controller \
  --set region=us-east-1 \
  --set vpcId=vpc-0bf8c5f8034b87a39

# verify
kubectl get deployment -n kube-system aws-load-balancer-controller
```

4. Add Helm Repositories

```
# Add the Helm Repos
helm repo add prometheus-community https://prometheus-community.github.io/helm-charts
helm repo add grafana https://grafana.github.io/helm-charts
helm repo update

# Create a Namespace
kubectl create namespace monitoring
```

5. EBS CSI Driver Setup

```
# Adding ebs-csi driver
aws eks create-addon \
  --cluster-name devops-monitoring \
  --addon-name aws-ebs-csi-driver \
  --service-account-role-arn arn:aws:iam::264152483755:role/AmazonEKS_EBS_CSI_DriverRole \
  --region us-east-1

# Create the IAM Role for EBS CSI
eksctl create iamserviceaccount \
  --name ebs-csi-controller-sa \
  --namespace kube-system \
  --cluster devops-monitoring \
  --role-name AmazonEKS_EBS_CSI_DriverRole \
  --attach-policy-arn arn:aws:iam::aws:policy/service-role/AmazonEBSCSIDriverPolicy \
  --approve \
  --override-existing-serviceaccounts

# Check / Set a Default StorageClass
kubectl patch storageclass gp2 -p '{"metadata": {"annotations":{"storageclass.kubernetes.io/
```

6. Install Prometheus

```
# Install Prometheus
helm upgrade --install prometheus prometheus-community/prometheus \
  --namespace monitoring \
  --set server.persistentVolume.enabled=true \
  --set server.persistentVolume.size=20Gi \
  --set server.persistentVolume.storageClass=gp2 \
  --set alertmanager.persistentVolume.enabled=true \
  --set alertmanager.persistentVolume.size=5Gi \
  --set alertmanager.persistentVolume.storageClass=gp2

# verify installation:
kubectl get pods -n monitoring
```

7. Install Grafana

```
# Install Grafana
helm upgrade grafana grafana/grafana \
  --namespace monitoring \
  --set adminPassword='admin123' \
  --set service.type=ClusterIP \
```

```
  --set ingress.enabled=false \
  --set persistence.enabled=true \
  --set persistence.size=10Gi \
  --set persistence.storageClassName=gp2 \
  --set grafana.ini.server.root_url='%(protocol)s://%(domain)s/' \
  --set grafana.ini.server.serve_from_sub_path=false

# Get Grafana Admin Password
kubectl get secret --namespace monitoring grafana -o jsonpath="{.data.admin-password}" | bas
```

8. Troubleshooting

```
# because of pvc issues
kubectl delete pvc -n monitoring prometheus-server

helm upgrade --install prometheus prometheus-community/prometheus \
  --namespace monitoring \
  --set server.persistentVolume.enabled=true \
  --set server.persistentVolume.size=20Gi \
  --set server.persistentVolume.storageClass=gp2 \
  --set alertmanager.persistentVolume.enabled=true \
  --set alertmanager.persistentVolume.size=5Gi \
  --set alertmanager.persistentVolume.storageClass=gp2

# To get services
kubectl get svc --namespace monitoring
```

9. IAM Policy Update

```
Edit IAM Policy AWSLoadBalancerControllerIAMPolicy and add below things:

{
  "Effect": "Allow",
  "Action": [
    "elasticloadbalancing:SetRulePriorities"
  ],
  "Resource": "*"
}
```

10. Ingress Configuration

```
# apply
kubectl apply -f grafana-ingress.yaml
kubectl apply -f prometheus-ingress.yaml

# verify
kubectl get ingress -n monitoring
```

11. Ingress YAML Files

Grafana Ingress (grafana-ingress.yaml):

```
apiVersion: networking.k8s.io/v1
kind: Ingress
metadata:
  name: grafana-ingress
  namespace: monitoring
  annotations:
    alb.ingress.kubernetes.io/scheme: internet-facing
    alb.ingress.kubernetes.io/target-type: ip
spec:
```

```yaml
  ingressClassName: alb
  rules:
    - http:
        paths:
          - path: /
            pathType: Prefix
            backend:
              service:
                name: grafana
                port:
                  number: 80
```

Prometheus Ingress (prometheus-ingress.yaml):

```yaml
apiVersion: networking.k8s.io/v1
kind: Ingress
metadata:
  name: prometheus-ingress
  namespace: monitoring
  annotations:
    alb.ingress.kubernetes.io/scheme: internet-facing
    alb.ingress.kubernetes.io/target-type: ip
spec:
  ingressClassName: alb
  rules:
    - http:
        paths:
          - path: /
            pathType: Prefix
            backend:
              service:
                name: prometheus-server
                port:
                  number: 80
```

This guide provides step-by-step instructions for setting up a complete EKS monitoring stack with Prometheus and Grafana. Make sure to replace the account ID and VPC ID with your actual values before running the commands.